

$$\begin{bmatrix} 1 & -2 & -6 & 11 \\ 0 & 3 & 7 & -5 \\ 1 & -2 & 5 & 9 \end{bmatrix} \xrightarrow[\text{-Row 1}]{\text{Row 3} = \text{Row 3}} \begin{bmatrix} 1 & -2 & -6 & 11 \\ 0 & 3 & 7 & -5 \\ 0 & 0 & 11 & -2 \end{bmatrix}$$

Verification of whether final matrix is in echelon form

1. There are no ^{zero} rows in the matrix $\therefore$ The condition that all non-zero rows must be above all zero rows is vacuously true.
 2. Row 1 Leading Entry = 1 in Column 1
Row 2 Leading Entry = 3 in Column 2
Row 3 Leading Entry = 11 in Column 3
 3. Row 2 Column 1 = Row 3 Column 1 = 0
Row 3 Column 2 = 0
- $\therefore$ The condition that each pivot column has only 0's below the pivot position is True.

The matrix $\begin{bmatrix} 1 & -2 & -6 & 11 \\ 0 & 3 & 7 & -5 \\ 0 & 0 & 11 & -2 \end{bmatrix}$ is in echelon form.

There are no rows in the matrix of the form $[0 \ 0 \ \dots \ 0 \ b]$ where $b \neq 0$. $\therefore$ The system is consistent, i.e. there exists a solution set. (Theorem 2)

$\therefore b = \begin{bmatrix} 11 \\ -5 \\ 9 \end{bmatrix}$ is a linear combination of the vectors formed from the columns of the matrix $A = \begin{bmatrix} 1 & -2 & -6 \\ 0 & 3 & 7 \\ 1 & -2 & 5 \end{bmatrix}$ $\square$

15. $v_1 = \begin{bmatrix} 7 \\ -6 \end{bmatrix}$, $v_2 = \begin{bmatrix} -5 \\ 3 \\ 0 \end{bmatrix}$. List five vectors in $\text{Span}\{v_1, v_2\}$

Solution: By definition, a vector b is in the Span of v_1, v_2 if b is a linear combination of v_1, v_2 . i.e. $\exists$ some $x, y \in \mathbb{R}$ s.t. $xv_1 + yv_2 = b$.

1) $x=1, y=1$

$$1 \begin{bmatrix} 7 \\ -6 \end{bmatrix} + 1 \begin{bmatrix} -5 \\ 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \times 7 \\ 1 \times 1 \\ 1 \times -6 \end{bmatrix} + \begin{bmatrix} 1 \times -5 \\ 1 \times 3 \\ 1 \times 0 \end{bmatrix} \quad (\text{By scalar multiplication})$$

$$= \begin{bmatrix} 7 + (-5) \\ 1 + 3 \\ -6 + 0 \end{bmatrix} \quad (\text{Vector addition}) = \begin{bmatrix} 2 \\ 4 \\ -6 \end{bmatrix}$$